

The interaction of atoms with radiation



To describe the interaction of a two-level atom with radiation we shall use a *semiclassical* treatment, i.e. the radiation is treated as a classical electric field but we use quantum mechanics to treat the atom. We shall calculate the effect of an oscillating electric field on the atom from first principles and show that this is equivalent to the usual time-dependent perturbation theory (TDPT) summarised by the golden rule (as mentioned in Section 2.2). The golden rule only gives the steady-state transition rate and therefore does not describe adequately spectroscopy experiments with highly monochromatic radiation, e.g. radio-frequency radiation, microwaves or laser light, in which the amplitudes of the quantum states evolve coherently in time. In such experiments the damping time may be less than the total measurement time so that the atoms never reach the steady state.

From the theory of the interaction with radiation, we will be able to find the conditions for which the equations reduce to a set of rate equations that describe the populations of the atomic energy levels (with a steady-state solution). In particular, for an atom illuminated by broadband radiation, this approach allows us to make a connection with Einstein’s treatment of radiation that was presented in Chapter 1; we shall find the Einstein B coefficient in terms of the matrix element for the transition. Then we can use the relation between A_{21} and B_{21} to calculate the spontaneous decay rate of the upper level. Finally, we shall study the roles of natural broadening and Doppler broadening in the absorption of radiation by atoms, and derive some results needed in later chapters such as the a.c. Stark shift.

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7.1 Setting up the equations

We start from the time-dependent Schrödinger equation¹

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi. \tag{7.1}$$

The Hamiltonian has two parts,

$$H = H_0 + H_I(t). \tag{7.2}$$

That part of the Hamiltonian that depends on time, $H_I(t)$, describes the interaction with the oscillating electric field that perturbs the eigen-

¹The operators do not have hats so $H \equiv \hat{H}$, as previously.

functions of H_0 ; the unperturbed eigenvalues and eigenfunctions of H_0 are just the atomic energy levels and wavefunctions that we found in previous chapters. We write the wavefunction for the level with energy E_n as

$$\Psi_n(\mathbf{r}, t) = \psi_n(\mathbf{r}) e^{-iE_n t/\hbar}. \quad (7.3)$$

For a system with only two levels, the spatial wavefunctions satisfy

$$\begin{aligned} H_0 \psi_1(\mathbf{r}) &= E_1 \psi_1(\mathbf{r}), \\ H_0 \psi_2(\mathbf{r}) &= E_2 \psi_2(\mathbf{r}). \end{aligned} \quad (7.4)$$

These atomic wavefunctions are not stationary states of the full Hamiltonian, $H_0 + H_I(t)$, but the wavefunction at any instant of time can be expressed in terms of them as follows:

$$\Psi(\mathbf{r}, t) = c_1(t) \psi_1(\mathbf{r}) e^{-iE_1 t/\hbar} + c_2(t) \psi_2(\mathbf{r}) e^{-iE_2 t/\hbar}, \quad (7.5)$$

or, in concise Dirac ket notation (shortening $c_1(t)$ to c_1 , $\omega_1 = E_1/\hbar$, etc.),

$$\Psi(\mathbf{r}, t) = c_1 |1\rangle e^{-i\omega_1 t} + c_2 |2\rangle e^{-i\omega_2 t}. \quad (7.6)$$

Normalisation requires that the two time-dependent coefficients satisfy

$$|c_1|^2 + |c_2|^2 = 1. \quad (7.7)$$

7.1.1 Perturbation by an oscillating electric field

The oscillating electric field $\mathbf{E} = \mathbf{E}_0 \cos(\omega t)$ of electromagnetic radiation produces a perturbation described by the Hamiltonian

$$H_I(t) = e\mathbf{r} \cdot \mathbf{E}_0 \cos(\omega t). \quad (7.8)$$

This corresponds to the energy of an electric dipole $-e\mathbf{r}$ in the electric field, where $\mathbf{r}$ is the position of the electron with respect to the atom's centre of mass.² Note that we have assumed that the electric dipole moment arises from a single electron but the treatment can easily be generalised by summing over all of the atom's electrons. The interaction mixes the two states with energies E_1 and E_2 . Substitution of eqn 7.6 into the time-dependent Schrödinger eqn 7.1 leads to

$$i\dot{c}_1 = \Omega \cos(\omega t) e^{-i\omega_0 t} c_2, \quad (7.9)$$

$$i\dot{c}_2 = \Omega^* \cos(\omega t) e^{i\omega_0 t} c_1, \quad (7.10)$$

where $\omega_0 = (E_2 - E_1)/\hbar$ and the Rabi frequency Ω is defined by

$$\Omega = \frac{\langle 1 | e\mathbf{r} \cdot \mathbf{E}_0 | 2 \rangle}{\hbar} = \frac{e}{\hbar} \int \psi_1^*(r) \mathbf{r} \cdot \mathbf{E}_0 \psi_2(r) d^3\mathbf{r}. \quad (7.11)$$

The electric field has almost uniform amplitude over the atomic wavefunction so we take the amplitude $|\mathbf{E}_0|$ outside the integral.³ Thus, for radiation linearly polarized along the x -axis, $\mathbf{E} = |\mathbf{E}_0| \hat{\mathbf{e}}_x \cos(\omega t)$, we obtain⁴

²Note that $\mathbf{E}_0 \cos(\omega t)$ is not replaced by a complex quantity $\mathbf{E}_0 e^{-i\omega t}$, because a complex convention is built into quantum mechanics and we must not confuse one thing with another.

³This *dipole* approximation holds when the radiation has a wavelength greater than the size of the atom, i.e. $\lambda \gg a_0$, as discussed in Section 2.2.

⁴Section 2.2 on selection rules shows how to treat other polarizations.

$$\Omega = \frac{eX_{12}|\mathbf{E}_0|}{\hbar}, \quad (7.12)$$

where

$$X_{12} = \langle 1|x|2 \rangle. \quad (7.13)$$

To solve the coupled differential equations for $c_1(t)$ and $c_2(t)$ we need to make further approximations.

7.1.2 The rotating-wave approximation

When all the population starts in the lower level, $c_1(0) = 1$ and $c_2(0) = 0$, integration of eqns 7.9 and 7.10 leads to

$$\begin{aligned} c_1(t) &= 1, \\ c_2(t) &= \frac{\Omega^*}{2} \left\{ \frac{1 - \exp[i(\omega_0 + \omega)t]}{\omega_0 + \omega} + \frac{1 - \exp[i(\omega_0 - \omega)t]}{\omega_0 - \omega} \right\}. \end{aligned} \quad (7.14)$$

This gives a reasonable first-order approximation while $c_2(t)$ remains small. For most cases of interest, the radiation has a frequency close to the atomic resonance at ω_0 so the magnitude of the detuning is small, $|\omega_0 - \omega| \ll \omega_0$, and hence $\omega_0 + \omega \sim 2\omega_0$. Therefore we can neglect the term with denominator $\omega_0 + \omega$ inside the curly brackets. This is the **rotating-wave approximation**.⁵ The modulus-squared of the co-rotating term gives the probability of finding the atom in the upper state at time t as

$$|c_2(t)|^2 = \left| \Omega \frac{\sin\{(\omega_0 - \omega)t/2\}}{\omega_0 - \omega} \right|^2, \quad (7.15)$$

or, in terms of the variable $x = (\omega - \omega_0)t/2$,

$$|c_2(t)|^2 = \frac{1}{4} |\Omega|^2 t^2 \frac{\sin^2 x}{x^2}. \quad (7.16)$$

The sinc function $(\sin x)/x$ has a maximum at $x = 0$, and the first minimum occurs at $x = \pi$ or $\omega_0 - \omega = \pm 2\pi/t$, as illustrated in Fig. 7.1; the frequency spread decreases as the interaction time t increases.

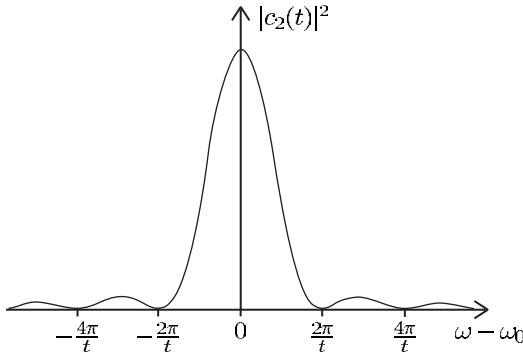


Fig. 7.1 The excitation probability function of the radiation frequency has a maximum at the atomic resonance. The line width is inversely proportional to the interaction time. The function sinc^2 also describes the Fraunhofer diffraction of light passing through a single slit—the diffraction angle decreases as the width of the aperture increases. The mathematical correspondence between these two situations has a natural explanation in terms of Fourier transforms.

⁵This is not true for the interaction of atoms with radiation at $10.6 \mu\text{m}$ from a CO_2 laser. This laser radiation has a frequency closer to d.c. than to the resonance frequency of the atoms, e.g. for rubidium with a resonance transition in the near infra-red (780 nm) we find that $\omega_0 \simeq 15\omega$, hence $\omega_0 + \omega \simeq \omega_0 - \omega$. Thus the counter-rotating term must be kept. The quasi-electrostatic traps (QUEST) formed by such long wavelength laser beams are a form of the dipole-force traps described in Chapter 10.

7.2 The Einstein B coefficients

In the previous section we found the effect of an electric field $\mathbf{E}_0 \cos(\omega t)$ on the atom. To relate this to Einstein's treatment of the interaction with broadband radiation we consider what happens with radiation of energy density $\rho(\omega)$ in the frequency interval ω to $\omega + d\omega$. This produces an electric field of amplitude $E_0(\omega)$ given by $\rho(\omega) d\omega = \epsilon_0 E_0^2(\omega)/2$. For this narrow (almost monochromatic) slice of the broad distribution eqn 7.12 gives

$$|\Omega|^2 = \left| \frac{eX_{12}E_0(\omega)}{\hbar} \right|^2 = \frac{e^2 |X_{12}|^2}{\hbar^2} \frac{2\rho(\omega) d\omega}{\epsilon_0}. \quad (7.17)$$

Integration of eqn 7.15 over frequency gives the excitation probability for the broadband radiation as

$$|c_2(t)|^2 = \frac{2e^2 |X_{12}|^2}{\epsilon_0 \hbar^2} \int_{\omega_0 - \Delta/2}^{\omega_0 + \Delta/2} \rho(\omega) \frac{\sin^2\{(\omega_0 - \omega)t/2\}}{(\omega_0 - \omega)^2} d\omega. \quad (7.18)$$

We integrate the squares of the amplitudes, rather than taking the square of the total amplitude, since contributions at different frequencies do not interfere.⁶ The range of integration Δ must be large compared to the extent of the sinc function, but this is easily fulfilled since as time increases this function becomes sharply peaked at ω_0 . (In the limit $t \rightarrow \infty$ it becomes a Dirac delta function—see Loudon (2000).) Over the small range about ω_0 where the sinc function has an appreciable value, a smooth function like $\rho(\omega)$ varies little, so we take $\rho(\omega_0)$ outside the integral. A change of variable to $x = (\omega - \omega_0)t/2$, as in eqn 7.16, leads to

$$|c_2(t)|^2 \simeq \frac{2e^2 |X_{12}|^2}{\epsilon_0 \hbar^2} \rho(\omega_0) \times \frac{t}{2} \int_{-\phi}^{+\phi} \frac{\sin^2 x}{x^2} dx. \quad (7.19)$$

The integration has limits of $x = \pm\phi = \pm\Delta t/4 \gg \pi$ and the integral approximates closely to $\int_{-\infty}^{\infty} x^{-2} \sin^2 x dx = \pi$. We make this assumption of a long interaction to find the steady-state excitation rate for broadband radiation. The probability of transition from level 1 to 2 increases linearly with time corresponding to a transition rate of

$$R_{12} = \frac{|c_2(t)|^2}{t} = \frac{\pi e^2 |X_{12}|^2}{\epsilon_0 \hbar^2} \rho(\omega_0). \quad (7.20)$$

Comparison with the upward rate $B_{12}\rho(\omega)$ in Einstein's treatment of radiation (eqn 1.25) shows that

$$B_{12} = \frac{\pi e^2 |D_{12}|^2}{3\epsilon_0 \hbar^2}, \quad (7.21)$$

where $|X_{12}|^2 \rightarrow |D_{12}|^2/3$, and D_{12} is the magnitude of the vector

$$\mathbf{D}_{12} = \langle 1 | \mathbf{r} | 2 \rangle \equiv \int \psi_1^* \mathbf{r} \psi_2 d^3\mathbf{r}. \quad (7.22)$$

⁶As in optics experiments with broadband light, it is intensities that are summed, e.g. the formation of white-light fringes in the Michelson interferometer.

The factor of $1/3$ arises from averaging of $\mathbf{D} \cdot \hat{\mathbf{e}}_{\text{rad}}$, where $\hat{\mathbf{e}}_{\text{rad}}$ is a unit vector along the electric field, over all the possible spatial orientations of the atom (see Exercise 7.6). The relation between A_{21} and B_{12} in eqn 1.32 leads to

$$A_{21} = \frac{g_1}{g_2} \frac{4\alpha}{3c^2} \times \omega^3 |D_{12}|^2, \quad (7.23)$$

where $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ is the fine-structure constant. The matrix element between the initial and final states in eqn 7.22 depends on an integral involving the electronic wavefunctions of the atom so, as emphasised previously, the Einstein coefficients are properties of the atom. For a typical allowed transition the matrix element has an approximate value of $D_{12} \simeq 3a_0$ (this can be calculated analytically for hydrogenic systems). Using this estimate of D_{12} in the equation gives $A_{21} \simeq 2\pi \times 10^7 \text{ s}^{-1}$ for a transition of wavelength $\lambda = 6 \times 10^{-7} \text{ m}$ and $g_1 = g_2 = 1$. Although we have not given a physical explanation of spontaneous emission, Einstein's argument allows us to calculate its rate; he obtained the relation between A_{21} and B_{21} and we have used TDPT to determine B_{21} from the atomic wavefunctions. For a two-level atom with an allowed transition between the levels the quantum mechanical result corresponds closely to the treatment of radiative decay using classical electromagnetism in Section 1.6.

7.3 Interaction with monochromatic radiation

The derivation of eqns 7.14 assumed that the monochromatic radiation perturbed the atom only weakly so that most of the population stayed in the initial state. We shall now find a solution without assuming a weak field. We write eqn 7.9 as

$$i\dot{c}_1 = c_2 \left\{ e^{i(\omega - \omega_0)t} + e^{-i(\omega + \omega_0)t} \right\} \frac{\Omega}{2}, \quad (7.24)$$

and similarly for eqn 7.10. The term with $(\omega + \omega_0)t$ oscillates very fast and therefore averages to zero over any reasonable interaction time—this is the rotating-wave approximation (Section 7.1.2) and it leads to

$$\begin{aligned} i\dot{c}_1 &= c_2 e^{i(\omega - \omega_0)t} \frac{\Omega}{2}, \\ i\dot{c}_2 &= c_1 e^{-i(\omega - \omega_0)t} \frac{\Omega^*}{2}. \end{aligned} \quad (7.25)$$

These combine to give

$$\frac{d^2 c_2}{dt^2} + i(\omega - \omega_0) \frac{dc_2}{dt} + \left| \frac{\Omega}{2} \right|^2 c_2 = 0. \quad (7.26)$$

The solution of this second-order differential equation for the initial conditions $c_1(0) = 1$ and $c_2(0) = 0$ gives the probability of being in the upper state as⁷

⁷For transitions between bound states the frequency Ω is real, so $|\Omega|^2 = \Omega^2$.

$$|c_2(t)|^2 = \frac{\Omega^2}{W^2} \sin^2\left(\frac{Wt}{2}\right), \quad (7.27)$$

where

$$W^2 = \Omega^2 + (\omega - \omega_0)^2. \quad (7.28)$$

At resonance $\omega = \omega_0$ and $W = \Omega$, so

$$|c_2(t)|^2 = \sin^2\left(\frac{\Omega t}{2}\right). \quad (7.29)$$

⁸This is partly a consequence of the dependence on ω^3 in eqn 7.23, but also because the **magnetic dipole** transitions have smaller matrix elements than electric dipole transitions. For electric dipole transitions in the optical region spontaneous emission washes out the Rabi oscillations on a time-scale of tens of nanoseconds ($\tau = 1/A_{21}$, assuming that the predominant decay is from 2 to 1, and we estimated A_{12} above). Nevertheless, experimenters have observed coherent oscillations by driving the transition with intense laser radiation to give a high Rabi frequency ($\Omega\tau > 1$).

The population oscillates between the two levels. When $\Omega t = \pi$ all the population has gone from level 1 into the upper state, $|c_2(t)|^2 = 1$, and when $\Omega t = 2\pi$ the atom has returned to the lower state. This behaviour is completely different from that of a two-level system governed by rate equations where the populations tend to become equal as the excitation rate increases and population inversion cannot occur. These **Rabi oscillations** between the two levels are readily observed in radio-frequency spectroscopy, e.g. for transitions between Zeeman or hyperfine states. Radio-frequency and microwave transitions have negligible spontaneous emission so that, in most cases, the atoms evolve coherently.⁸

7.3.1 The concepts of π -pulses and $\pi/2$ -pulses

A pulse of resonant radiation that has a duration of $t_\pi = \pi/\Omega$ is called a π -pulse and from eqn 7.29 we see that $\Omega t = \pi$ results in the complete transfer of population from one state to the other, e.g. an atom initially in $|1\rangle$ ends up in $|2\rangle$ after the pulse. This contrasts with illumination by broadband radiation where the populations (per state) become equal as the energy density $\rho(\omega)$ increases. More precisely, a π -pulse swaps the states in a superposition:⁹

$$c_1|1\rangle + c_2|2\rangle \rightarrow -i\{c_1|2\rangle + c_2|1\rangle\}. \quad (7.30)$$

This swap operation is sometimes also expressed as $|1\rangle \leftrightarrow |2\rangle$, but the factor of $-i$ is important in atom interferometry, as shown in Exercise 7.3.

Interferometry experiments also use $\pi/2$ -pulses that have half the duration of a π -pulse (for the same Rabi frequency Ω). For an atom initially in state $|1\rangle$, the $\pi/2$ -pulse puts its wavefunction into a superposition of states $|1\rangle$ and $|2\rangle$ with equal amplitudes (see Exercise 7.3).

7.3.2 The Bloch vector and Bloch sphere

In this section we find the electric dipole moment induced on a atom by radiation, and introduce a very powerful way of describing the behaviour of two-level systems by the Bloch vector. We assume that the electric field is along $\hat{\mathbf{e}}_x$, as in eqn 7.12. The component of the dipole along this direction is given by the expectation value

$$-eD_x(t) = -\int \Psi^\dagger(t) ex \Psi(t) d^3\mathbf{r}. \quad (7.31)$$

⁹This can be shown by solving eqns 7.25 (and 7.26) for $\omega = \omega_0$.

Using eqn 7.5 for $\Psi(t)$ gives this dipole moment of the atom as

$$\begin{aligned} D_x(t) &= \int (c_1 e^{-i\omega_1 t} \psi_1 + c_2 e^{-i\omega_2 t} \psi_2)^* x (c_1 e^{-i\omega_1 t} \psi_1 + c_2 e^{-i\omega_2 t} \psi_2) d^3 \mathbf{r} \\ &= c_2^* c_1 X_{21} e^{i\omega_0 t} + c_1^* c_2 X_{12} e^{-i\omega_0 t}. \end{aligned} \quad (7.32)$$

Here $\omega_0 = \omega_2 - \omega_1$. The dipole moment is a real quantity since from eqn 7.13 we see that $X_{21} = (X_{12})^*$, and also $X_{11} = X_{22} = 0$. To calculate this dipole moment induced by the applied field we need to know the bilinear quantities $c_1^* c_2$ and $c_2^* c_1$. These are some of the elements of the density matrix¹⁰

$$|\Psi\rangle\langle\Psi| = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \begin{pmatrix} c_1^* & c_2^* \end{pmatrix} = \begin{pmatrix} |c_1|^2 & c_1 c_2^* \\ c_2 c_1^* & |c_2|^2 \end{pmatrix} = \begin{pmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{pmatrix}. \quad (7.33)$$

Off-diagonal elements of the density matrix are called **coherences** and they represent the response of the system at the driving frequency (eqn 7.32). The diagonal elements $|c_1|^2$ and $|c_2|^2$ are the populations. We define the new variables

$$\tilde{c}_1 = c_1 e^{-i\delta t/2}, \quad (7.34)$$

$$\tilde{c}_2 = c_2 e^{i\delta t/2}, \quad (7.35)$$

where $\delta = \omega - \omega_0$ is the detuning of the radiation from the atomic resonance. This transformation does not affect the populations ($\tilde{\rho}_{11} = \rho_{11}$ and $\tilde{\rho}_{22} = \rho_{22}$) but the coherences become $\tilde{\rho}_{12} = \rho_{12} \exp(-i\delta t)$ and $\tilde{\rho}_{21} = \rho_{21} \exp(i\delta t) = (\tilde{\rho}_{12})^*$. In terms of these coherences the dipole moment is¹¹

$$\begin{aligned} -eD_x(t) &= -eX_{12} \{\rho_{12} e^{i\omega_0 t} + \rho_{21} e^{-i\omega_0 t}\} = -eX_{12} \{\tilde{\rho}_{12} e^{i\omega t} + \tilde{\rho}_{21} e^{-i\omega t}\} \\ &= -eX_{12} (u \cos \omega t - v \sin \omega t). \end{aligned} \quad (7.36)$$

The coherences $\tilde{\rho}_{12}$ and $\tilde{\rho}_{21}$ give the response of the atom at ω , the (angular) frequency of the applied field. The real and imaginary parts of $\tilde{\rho}_{12}$ (multiplied by 2) are:

$$\begin{aligned} u &= \tilde{\rho}_{12} + \tilde{\rho}_{21}, \\ v &= -i(\tilde{\rho}_{12} - \tilde{\rho}_{21}). \end{aligned} \quad (7.37)$$

In eqn 7.36 we see that u and v are the in-phase and quadrature components of the dipole in a frame rotating at ω . To find expressions for $\tilde{\rho}_{12}$, $\tilde{\rho}_{21}$ and ρ_{22} , and hence u and v , we start by writing eqns 7.25 for c_1 and c_2 in terms of δ as follows:¹²

$$i\dot{c}_1 = c_2 e^{i\delta t} \frac{\Omega}{2}, \quad (7.38)$$

$$i\dot{c}_2 = c_1 e^{-i\delta t} \frac{\Omega}{2}. \quad (7.39)$$

Differentiation of eqn 7.34 yields¹³

¹⁰This is the outer product of $|\Psi\rangle$ and its Hermitian conjugate $\langle\Psi| \equiv |\Psi\rangle^\dagger$, the transposed conjugate of the matrix representing $|\Psi\rangle$. This way of writing the information about the two levels is an extremely useful formalism for treating quantum systems. However, we have no need to digress into the theory of density matrices here and we simply adopt it as a convenient notation.

¹¹We assume that X_{12} is real. This is true for transitions between two bound states of the atom—the radial wavefunctions are real and the discussion of selection rules shows that the integration over the angular momentum eigenfunctions also gives a real contribution to the matrix element—the integral over ϕ is zero unless the terms containing powers of $\exp(-i\phi)$ cancel.

¹²All the steps in this lengthy procedure cannot be written down here but enough information is given for meticulous readers to fill in the gaps.

¹³Spontaneous decay is ignored here—this section deals only with coherent evolution of the states.

$$\dot{\tilde{c}}_1 = \dot{c}_1 e^{-i\delta t/2} - \frac{i\delta}{2} c_1 e^{-i\delta t/2}. \quad (7.40)$$

Multiplication by i and the use of eqns 7.38, 7.34 and 7.35 yields an equation for $\dot{\tilde{c}}_1$ (and similarly we can obtain $\dot{\tilde{c}}_2$ from eqn 7.39):

$$\begin{aligned} i\dot{\tilde{c}}_1 &= \frac{1}{2}(\delta \tilde{c}_1 + \Omega \tilde{c}_2), \\ i\dot{\tilde{c}}_2 &= \frac{1}{2}(\Omega \tilde{c}_1 - \delta \tilde{c}_2). \end{aligned} \quad (7.41)$$

From these we find that the time derivatives $\dot{\tilde{\rho}}_{12} = \tilde{c}_1 \dot{\tilde{c}}_2^* + \dot{\tilde{c}}_1 \tilde{c}_2^*$, etc. are

$$\begin{aligned} \frac{d\tilde{\rho}_{12}}{dt} &= \left(\frac{d\tilde{\rho}_{21}}{dt} \right)^* = -i\delta \tilde{\rho}_{12} + \frac{i\Omega}{2}(\rho_{11} - \rho_{22}), \\ \frac{d\rho_{22}}{dt} &= -\frac{d\rho_{11}}{dt} = \frac{i\Omega}{2}(\tilde{\rho}_{21} - \tilde{\rho}_{12}). \end{aligned} \quad (7.42)$$

The last equation is consistent with normalisation in eqn 7.7, i.e.

$$\rho_{22} + \rho_{11} = 1. \quad (7.43)$$

In terms of u and v in eqns 7.37 these equations become

$$\begin{aligned} \dot{u} &= \delta v, \\ \dot{v} &= -\delta u + \Omega(\rho_{11} - \rho_{22}), \\ \dot{\rho}_{22} &= \frac{\Omega v}{2}. \end{aligned} \quad (7.44)$$

¹⁴This is appropriate for calculating absorption. Alternatively, $\rho_{22} - \rho_{11}$ could be chosen as a variable—this population inversion determines the gain in lasers.

We can write the population difference $\rho_{11} - \rho_{22}$ as¹⁴

$$w = \rho_{11} - \rho_{22}, \quad (7.45)$$

so that finally we get the following compact set of equations:

$$\begin{aligned} \dot{u} &= \delta v, \\ \dot{v} &= -\delta u + \Omega w, \\ \dot{w} &= -\Omega v. \end{aligned} \quad (7.46)$$

These eqns 7.46 can be written in vector notation as:

$$\dot{\mathbf{R}} = \mathbf{R} \times (\Omega \hat{\mathbf{e}}_1 + \delta \hat{\mathbf{e}}_3) = \mathbf{R} \times \mathbf{W}, \quad (7.47)$$

by taking u , v and w as the components of the Bloch vector

$$\mathbf{R} = u \hat{\mathbf{e}}_1 + v \hat{\mathbf{e}}_2 + w \hat{\mathbf{e}}_3, \quad (7.48)$$

and defining the vector

$$\mathbf{W} = \Omega \hat{\mathbf{e}}_1 + \delta \hat{\mathbf{e}}_3 \quad (7.49)$$

that has magnitude $W = \sqrt{\Omega^2 + \delta^2}$ (cf. eqn 7.28). The cross-product of the two vectors in eqn 7.47 is orthogonal to both $\mathbf{R}$ and $\mathbf{W}$. This implies

that $\dot{\mathbf{R}} \cdot \mathbf{R} = 0$ so $|\mathbf{R}|^2$ is constant—it is straightforward¹⁵ to show that this constant is unity so that $|\mathbf{R}|^2 = |u|^2 + |v|^2 + |w|^2 = 1$. The Bloch vector corresponds to the position vector of points on the surface of a sphere with unit radius; this Bloch sphere is shown in Fig. 7.2.

It also follows from eqn 7.47 that $\mathbf{R} \cdot \mathbf{W} = 0$ and so $\mathbf{R} \cdot \mathbf{W} = RW \cos \theta$ is constant. For excitation with a fixed Rabi frequency and detuning the magnitude W is constant, and since R is also fixed the Bloch vector moves around a cone with θ constant, as illustrated in Fig. 7.2(d). In this case ρ_{22} varies as in eqn 7.27 and

$$w = 1 - 2\rho_{22} = 1 - \frac{2\Omega^2}{W^2} \sin^2 \left(\frac{Wt}{2} \right).$$

This motion of the Bloch vector for the state of the atom resembles that of a magnetic moment in a magnetic field,¹⁶ e.g. for adiabatic motion the

¹⁵ $|\mathbf{R}| = 1$ for the state $u = v = 0$, $w = 1$ and it always remains unity. This can also be proved by writing u , v and w in terms of $|c_1|^2$, $|c_2|^2$, etc.

¹⁶As described in Blundell (2001, Appendix G).

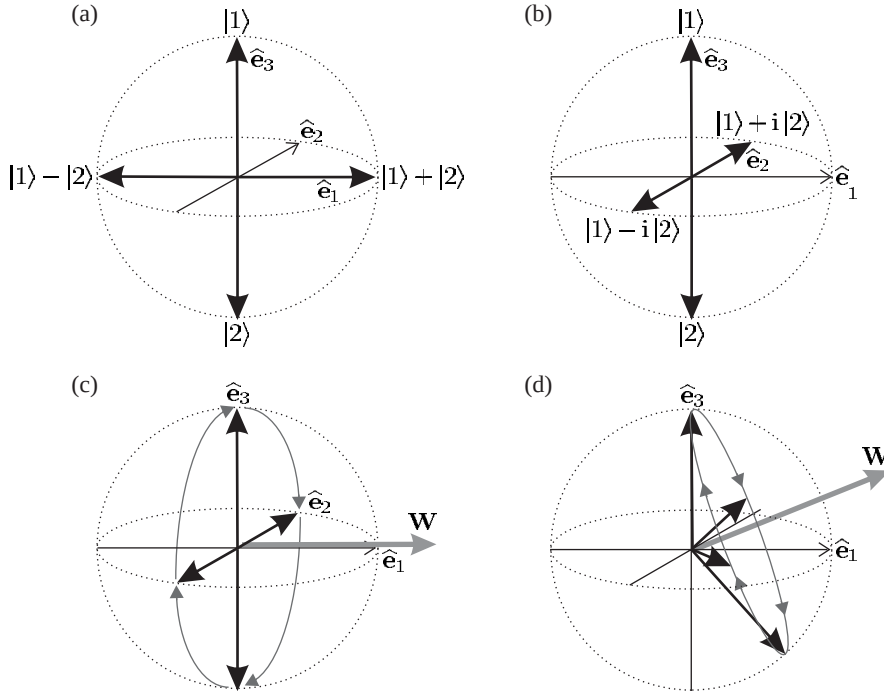


Fig. 7.2 The Bloch sphere. The position vectors of points on its surface represent the states of a two-level system (in Hilbert space). Examples of states are shown in (a) and (b). At the poles of the sphere the Bloch vector is $\mathbf{R} = w\hat{\mathbf{e}}_3$, with $w = \pm 1$ corresponding to the states $|1\rangle$ and $|2\rangle$, respectively. States that lie on the equator of the Bloch sphere have the form $\mathbf{R} = u\hat{\mathbf{e}}_1 + v\hat{\mathbf{e}}_2$, e.g. the states for which $\mathbf{R} = v\hat{\mathbf{e}}_2$ with $u = 0$ and $v = \pm 1$ are shown in (b) and these correspond to $(|1\rangle \pm i|2\rangle)/\sqrt{2}$, respectively (normalisation constants are not given in the figure for clarity). These examples illustrate an interesting property of this representation of quantum states, namely that diametrically-opposite states on the Bloch sphere are orthogonal. (c) The evolution of the Bloch vector for a system driven by a resonant field, i.e. $\delta = 0$ so that $\mathbf{W} = \Omega\hat{\mathbf{e}}_1$ in eqn 7.49. The evolution follows a great circle from $|1\rangle$ at the north pole to $|2\rangle$ at the south pole and back again, as described in Example 7.1. The Bloch vector remains perpendicular to $\mathbf{W}$. (d) When $\delta \neq 0$ the Bloch vector also has a fixed angle with respect to $\mathbf{W}$, since $\mathbf{R} \cdot \mathbf{W} = RW \cos \theta$ is constant, but θ is not equal to $\pi/2$. (This quantum mechanical description of the two-level atom is equivalent to that for a spin-1/2 system.)

energy $-\boldsymbol{\mu} \cdot \mathbf{B}$ is constant, and the magnetic moment precesses around the direction of the field $\mathbf{B} = B\hat{\mathbf{e}}_z$. In the Bloch description the fictitious magnetic field lies along $\mathbf{W}$ and the magnitude W in eqn 7.28 determines the precession rate.

Example 7.1 Resonant excitation ($\delta = 0$) gives $\mathbf{W} = \Omega\hat{\mathbf{e}}_1$ and $\mathbf{R}$ describes a cone about $\hat{\mathbf{e}}_1$. An important case is when all the population starts in level 1 so that initially $\mathbf{R} \cdot \hat{\mathbf{e}}_1 = 0$; in this case the Bloch vector rotates in the plane perpendicular to $\hat{\mathbf{e}}_1$ mapping out a great circle on the Bloch sphere, as drawn in Fig. 7.2(c). This motion corresponds to the Rabi oscillations (eqn 7.29). In this picture a $\pi/2$ -pulse rotates the Bloch vector through $\pi/2$ about $\hat{\mathbf{e}}_1$. A sequence of two $\pi/2$ -pulses gives a π -pulse that rotates the Bloch vector (clockwise) through π about $\hat{\mathbf{e}}_1$, e.g. $w = 1 \rightarrow w = -1$ and this represents the transfer of all the population from level 1 to 2.¹⁷ This is consistent with the more general statement given in eqn 7.30.

¹⁷In this particular example the final state is obvious by inspection, but clearly the same principles apply to other initial states, e.g. states of the form $\{|1\rangle + e^{i\phi}|2\rangle\}/\sqrt{2}$ that lie on the equator of the sphere. The Bloch sphere is indispensable for thinking about more complex pulse sequences, such as those used in nuclear magnetic resonance (NMR).

The very brief introduction to the Bloch sphere given in this section shows clearly that a two-level atom's response to radiation does not increase indefinitely with the driving field—beyond a certain point an increase in the applied field (or the interaction time) does not produce a larger dipole moment or change in population. This 'saturation' has important consequences and makes the two-level system different from a classical oscillator (where the dipole moment is proportional to the field, as will be shown in Section 7.5).

7.4 Ramsey fringes

The previous sections in this chapter have shown how to calculate the response of a two-level atom to radiation. In this section we shall apply this theory to radio-frequency spectroscopy, e.g. the method of magnetic resonance in an atomic beam described in Chapter 6. However, the same principles are important whenever line width is limited by the finite interaction time, both within atomic physics and more generally. In particular, we shall calculate what happens to an atom subjected to two pulses of radiation since such a double-pulse sequence has favourable properties for precision measurements.

An atom that interacts with a square pulse of radiation, i.e. an oscillating electric field of constant amplitude from time $t = 0$ to τ_p , and $E_0 = 0$ otherwise, has a probability of excitation as in eqn 7.15.¹⁸ This excitation probability is plotted in Fig. 7.1 as a function of the radiation's frequency detuning from the (angular) resonance frequency ω_0 . As stated below eqn 7.16, the frequency spread given by the first minimum of the sinc^2 function corresponds to a width¹⁹

$$\Delta f = \frac{\Delta\omega}{2\pi} = \frac{1}{\tau_p}. \quad (7.50)$$

The frequency spread is inversely proportional to the interaction time,²⁰

¹⁸This assumes weak excitation: $|c_2|^2 \ll 1$.

¹⁹This is not the FWHM but it is close enough for our purposes.

²⁰This expression is equivalent to eqn 6.40 that was used to calculate the line width for an atomic clock.

as expected from the Fourier transform relationship of the frequency and time domains.

We shall now consider what happens when an atom interacts with two separate pulses of radiation, from time $t = 0$ to τ_p and again from $t = T$ to $T + \tau_p$. Integration of eqn 7.10 with the initial condition $c_2 = 0$ at $t = 0$ yields

$$c_2(t) = \frac{\Omega^*}{2} \left\{ \frac{1 - \exp[i(\omega_0 - \omega)\tau_p]}{\omega_0 - \omega} + \exp[i(\omega_0 - \omega)T] \frac{1 - \exp[i(\omega_0 - \omega)\tau_p]}{\omega_0 - \omega} \right\}. \quad (7.51)$$

This is the amplitude excited to the upper level after both pulses ($t > T + \tau_p$). The first term in this expression is the amplitude arising from the first pulse and it equals the part of eqn 7.14 that remains after making the rotating-wave approximation.²¹ Within this approximation, interaction with the second pulse produces a similar term multiplied by a phase factor of $\exp[i(\omega_0 - \omega)T]$. Either of the pulses acting alone would affect the system in the same way, i.e. the same excitation probability $|c_2|^2$ as in eqn 7.15. When there are two pulses the amplitudes in the excited state interfere giving

$$\begin{aligned} |c_2|^2 &= \left| \Omega \frac{\sin\{(\omega_0 - \omega)\tau_p/2\}}{(\omega_0 - \omega)} \right|^2 \times |1 + \exp[i(\omega_0 - \omega)T]|^2 \\ &= \left| \frac{\Omega\tau_p}{2} \right|^2 \left[\frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2} \right]^2 \cos^2\left(\frac{\delta T}{2}\right), \end{aligned} \quad (7.52)$$

where $\delta = \omega - \omega_0$ is the frequency detuning. The double-pulse sequence produces a signal of the form shown in Fig. 7.3. These are called **Ramsey fringes** after Norman Ramsey and they have a very close similarity to the interference fringes seen in a Young's double-slit experiment in optics—Fraunhofer diffraction of light with wavevector k from two slits of width a and separation d leads to an intensity distribution as a function of angle θ given by²²

$$I = I_0 \cos^2\left(\frac{1}{2}kd\sin\theta\right) \text{sinc}^2\left(\frac{1}{2}ka\sin\theta\right). \quad (7.53)$$

The overall envelope proportional to sinc^2 comes from single-slit diffraction. The $\cos^2$ function determines the width of the central peak in both eqns 7.53 and 7.52.²³

For the atom excited by two pulses of radiation the excitation drops from the maximum value at $\omega = \omega_0$ to zero when $\delta T/2 = \pi/2$ (or to half the maximum at $\pi/4$); so the central peak has a width (FWHM) of $\Delta\omega = \pi/T$, or equivalently

$$\Delta f = \frac{1}{2T}. \quad (7.54)$$

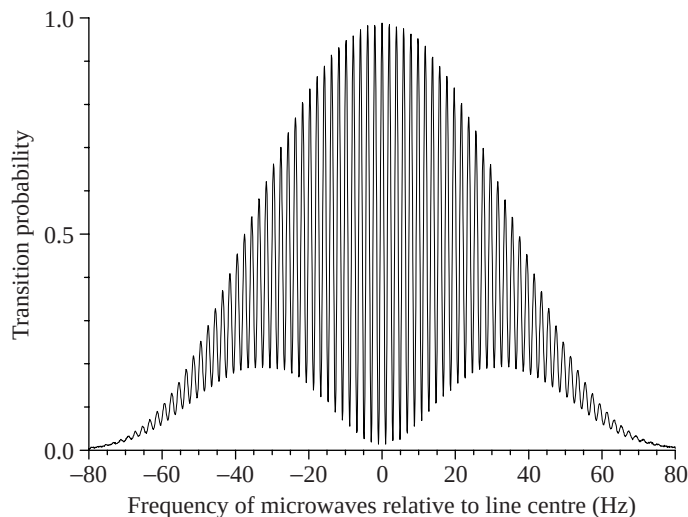
This shows that Ramsey fringes from two interactions separated by time T have half the width of the signal from a single long interaction of

²¹Neglecting terms with $\omega_0 + \omega$ in the denominator.

²²See Section 11.1 and Brooker (2003).

²³In both quantum mechanics and optics, the amplitudes of waves interfere constructively, or destructively, depending on their relative phase. Also, the calculation of Fraunhofer diffraction as a Fourier transform of amplitude in the plane of the object closely parallels the Fourier transform relationship between pulses in the time domain and the frequency response of the system.

Fig. 7.3 Ramsey fringes from an atomic fountain of caesium, showing the transition probability for the $F = 3$, $M_F = 0$ to $F' = 4$, $M_{F'} = 0$ transition versus the frequency of the microwave radiation in the interaction region. The height of the fountain is 31 cm, giving a fringe width just below 1 Hz (i.e. $\Delta f = 1/(2T) = 0.98$ Hz, see text)—the envelope of the fringes has a more complicated shape than that derived in the text, but this has little influence since, during operation as a frequency standard, the microwaves have a frequency very close to the centre which, by definition, corresponds to 9 192 631 770 Hz. This is real experimental data but the noise is not visible because the signal-to-noise ratio is about 1000 (near the centre), and with such an extremely high-quality signal the short-term stability of a microwave source referenced to the caesium transition is about 1×10^{-13} for 1 s of averaging. Courtesy of Dale Henderson, Krzysztof Szymaniec and Chalupczak Witold, National Physical Laboratory, Teddington, UK.



²⁴Ramsey originally introduced two separated interactions in an atomic-beam experiment to avoid the line broadening by inhomogeneous magnetic fields. A small phase difference between the two interaction regions just causes a phase shift of the fringes, whereas if the atom interacts with the radiation throughout a region where the field varies then the contributions from each part of the interaction region do not add in phase. The phasor description, commonly used in optics, gives a good way of thinking about this—Young’s fringes have a high contrast when the two slits of separation d are illuminated coherently, but to achieve the diffraction limit from a single wide slit of width d requires a good wavefront across the whole aperture.

²⁵In the Bloch sphere description this corresponds to the following path: the initial $\pi/2$ -pulse causes $\hat{e}_3 \rightarrow \hat{e}_2$, then the accumulated phase causes the state vector to move around the equator of the sphere to $-\hat{e}_2$, from whence the final $\pi/2$ -pulse takes the system back up to $\hat{e}_3$ (see Fig. 7.2). This formalism allows quantitative calculation of the final state for any type of pulse.

duration T (cf. eqn 7.50); also, it is often preferable to have two separated interaction regions, e.g. for measurements in an atomic fountain as described in Chapter 8.²⁴

In practice, microwave experiments use strong rather than weak excitation (as assumed above) to obtain the maximum signal, i.e. $|c_2|^2 \simeq 1$. This does not change the width of the Ramsey fringes, as shown by considering two $\pi/2$ -pulses separated by time T . If no phase shift accumulates between the two pulses they add together to act as a π -pulse that transfers all the population to the upper state—from the north to the south pole of the Bloch sphere, as shown in Fig. 7.2(c). But if a relative phase shift of π accrues during the time interval T then there is destructive interference between the amplitudes in the upper state produced by the two pulses.²⁵ Thus the first minimum from the central fringe occurs for $\delta T = \pi$, which is the condition that gave eqn 7.54, and so that equation remains accurate.

7.5 Radiative damping

This section shows how damping affects the coherent evolution of the Bloch vector described in the previous section. It is shown by analogy with the description of a classical dipole that a damping term should be introduced into eqns 7.46. Ultimately, such an argument by analogy is only a justification that the equations have an appropriate form rather

than a derivation, but this approach does give useful physical insight.

7.5.1 The damping of a classical dipole

Damping by spontaneous emission can be introduced into the quantum treatment of the two-level atom, in a physically reasonable way, by comparison with the damping of a classical system. To do this we first review the damped harmonic oscillator and express the classical equations in a suitable form. For a harmonic oscillator of natural frequency ω_0 , Newton's second law leads to the equation of motion

$$\ddot{x} + \beta\dot{x} + \omega_0^2 x = \frac{F(t)}{m} \cos \omega t. \quad (7.55)$$

The driving force has amplitude $F(t)$ that varies slowly compared to the oscillation at the driving frequency. (The friction force is $F_{\text{friction}} = -\alpha\dot{x}$ and $\beta = \alpha/m$, where m is the mass.) To solve this we look for a solution of the form

$$x = \mathcal{U}(t) \cos \omega t - \mathcal{V}(t) \sin \omega t. \quad (7.56)$$

This anticipates that most of the time dependence of the solution is an oscillation at frequency ω , and $\mathcal{U}$ is the component of the displacement in-phase with the force, and the quadrature component $\mathcal{V}$ has a phase lead²⁶ of $\pi/2$ with respect to $F \cos \omega t$.²⁷ Substitution of eqn 7.56 into eqn 7.55 and equating terms that depend on $\sin \omega t$ and $\cos \omega t$ gives

$$\begin{aligned} \dot{\mathcal{U}} &= (\omega - \omega_0) \mathcal{V} - \frac{\beta}{2} \mathcal{U}, \\ \dot{\mathcal{V}} &= -(\omega - \omega_0) \mathcal{U} - \frac{\beta}{2} \mathcal{V} - \frac{F(t)}{2m\omega}, \end{aligned} \quad (7.57)$$

respectively. The amplitudes $\mathcal{U}$ and $\mathcal{V}$ change in time as the amplitude of the force changes, but we assume that these changes occur slowly compared to the fast oscillation at ω . This **slowly-varying envelope approximation** has been used in the derivation of eqn 7.57, i.e. $\ddot{\mathcal{U}}$ and $\ddot{\mathcal{V}}$ have been neglected and $\dot{\mathcal{V}} \ll \omega \mathcal{V}$ (see Allen and Eberly 1975). By setting $\dot{\mathcal{U}} = \dot{\mathcal{V}} = 0$ we find the form of the solution that is a good approximation when the amplitudes and the force change slowly compared to the damping time of the system $1/\beta$:

$$\mathcal{U} = \frac{\omega_0 - \omega}{(\omega - \omega_0)^2 + (\beta/2)^2} \frac{F}{2m\omega}, \quad (7.58)$$

$$\mathcal{V} = \frac{-\beta/2}{(\omega - \omega_0)^2 + (\beta/2)^2} \frac{F}{2m\omega}. \quad (7.59)$$

The approximation $\omega^2 - \omega_0^2 = (\omega + \omega_0)(\omega - \omega_0) \simeq 2\omega(\omega - \omega_0)$ has been used so these expressions are only valid close to resonance²⁸—they give the wrong result for $\omega \simeq 0$. The phase is found from $\tan \phi = \mathcal{V}/\mathcal{U}$ (see Exercise 7.7). The phase lies in the range $\phi = 0$ to $-\pi$ for a force of constant amplitude.²⁹

²⁶A phase lead occurs when $\mathcal{V}(t) > 0$, since $-\sin \omega t = \cos(\omega t + \pi/2)$, and $\mathcal{V}(t) < 0$ corresponds to a phase lag.

²⁷This method of considering the components $\mathcal{U}$ and $\mathcal{V}$ is equivalent to the phasor description which is widely used in the theory of a.c. circuits (made from capacitors, inductors and resistors) to represent the phase lag, or lead, between the current and an applied voltage of the form $V_0 \cos \omega t$.

²⁸This is a very good approximation for optical transitions since typically $\beta/\omega_0 \simeq 10^{-6}$. The assumption of small damping is implicit in these equations and therefore the resonance frequency is very close to ω_0 .

²⁹It is well known from the study of the harmonic oscillator with damping that the mechanical response lags behind the driving. At low frequencies the system closely follows the driving force, but above the resonance, where $\omega > \omega_0$, the phase shift lies in the range $-\pi/2 < \phi < -\pi$.

The sum of the kinetic energy $\frac{1}{2} m \dot{x}^2$ and the potential energy $\frac{1}{2} m \omega_0^2 x^2$ gives the total energy $E = \frac{1}{2} m \omega^2 (\mathcal{U}^2 + \mathcal{V}^2)$ using the approximation $\omega_0^2 \simeq \omega^2$. This changes at the rate $\dot{E} = m \omega^2 (\mathcal{U} \dot{\mathcal{U}} + \mathcal{V} \dot{\mathcal{V}})$, and hence from eqns 7.57 for $\dot{\mathcal{U}}$ and $\dot{\mathcal{V}}$ we find that

$$\dot{E} = -\beta E - F \mathcal{V} \frac{\omega}{2}. \quad (7.60)$$

For no driving force ($F = 0$) the energy decays away. This is consistent with the complementary function of eqn 7.55 (the solution for $F = 0$) that gives the oscillator's transient response as³⁰

$$x = x_0 e^{-\beta t/2} \cos(\omega' t + \varphi). \quad (7.61)$$

Energy is proportional to the amplitude of the motion squared, hence the $\exp(-\beta t/2)$ dependence in eqn 7.61 becomes $E \propto \exp(-\beta t)$. The term $F \mathcal{V} \omega/2$ in eqn 7.60 is the rate at which the driving force does work on the oscillator; this can be seen from the following expression for power as the force times the velocity:

$$\overline{P} = \overline{F(t) \cos(\omega t) \dot{x}}. \quad (7.62)$$

The overlining indicates an average over many periods of the oscillation at ω , but the amplitude of the force $F(t)$ may vary (slowly) on a longer time-scale. Differentiation of eqn 7.56 gives the velocity as³¹

$$\dot{x} \simeq -\mathcal{U} \omega \sin \omega t - \mathcal{V} \omega \cos \omega t, \quad (7.63)$$

and only the cosine term contributes to the cycle-averaged power:³²

$$\overline{P} = -F(t) \mathcal{V} \frac{\omega}{2}. \quad (7.64)$$

This shows that absorption of energy arises from the quadrature component of the response $\mathcal{V}$.

In the classical model of an atom as an electron that undergoes simple harmonic motion the oscillating electric field of the incident radiation produces a force $F(t) = -e|\mathbf{E}_0| \cos \omega t$ on the electron. Each atom in the sample has an electric dipole moment of $D = -ex$ (along the direction of the applied field). The quadrature component of the dipole that gives absorption has a Lorentzian function of frequency as in eqn 7.59. The in-phase component of the dipole that determines the polarization of the medium and its refractive index (Fox 2001) has the frequency dependence given in eqn 7.58.³³

When any changes in the driving force occur slowly eqn 7.60 has the following quasi-steady-state solution:

$$E = \frac{|F \mathcal{V}| \omega}{2\beta}. \quad (7.65)$$

This shows that the energy of the classical oscillator increases linearly with the strength of the driving force, whereas in a two-level system the energy has an upper limit when all the atoms have been excited to the upper level.

³⁰For light damping ($\beta/\omega_0 \ll 1$) the decaying oscillations have angular frequency $\omega' = \{\omega_0^2 - \beta^2/4\}^{1/2} \simeq \omega_0$.

³¹We use the same slowly varying envelope approximation as for eqn 7.57, namely $\mathcal{V} \ll \omega \mathcal{V}$, etc.

³²This is normally a positive quantity since $r < 0$ (see eqn 7.59).

³³See Fig. 9.12.

7.5.2 The optical Bloch equations

A two-level atom has an energy proportional to the excited-state population, $E = \rho_{22}\hbar\omega_0$. By analogy with eqn 7.60 for the energy of a classical oscillator, we introduce a damping term into eqn 7.44 to give

$$\dot{\rho}_{22} = -\Gamma\rho_{22} + \frac{\Omega}{2}v. \quad (7.66)$$

In the absence of the driving term ($\Omega = 0$) this gives exponential decay of the population in level 2, i.e. $\rho_{22}(t) = \rho_{22}(0)\exp(-\Gamma t)$. In this analogy, between the quantum system and a classical oscillator, Γ corresponds to β . From eqns 7.57 we see that the coherences u and v have a damping factor of $\Gamma/2$ and eqns 7.46 become the **optical Bloch equations**³⁴

$$\begin{aligned} \dot{u} &= \delta v - \frac{\Gamma}{2}u, \\ \dot{v} &= -\delta u + \Omega w - \frac{\Gamma}{2}v, \\ \dot{w} &= -\Omega v - \Gamma(w - 1). \end{aligned} \quad (7.67)$$

For $\Omega = 0$ the population difference $w \rightarrow 1$. These optical Bloch equations describe the excitation of a two-level atom by radiation close to resonance for a transition that decays by spontaneous emission. There is not room here to explore all of the features of these equations and their many diverse and interesting applications; we shall concentrate on the steady-state solution that is established at times which are long compared to the lifetime of the upper level ($t \gg \Gamma^{-1}$), namely³⁵

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \frac{1}{\delta^2 + \Omega^2/2 + \Gamma^2/4} \begin{pmatrix} \Omega\delta \\ \Omega\Gamma/2 \\ \delta^2 + \Gamma^2/4 \end{pmatrix}. \quad (7.68)$$

These show that a strong driving field ($\Omega \rightarrow \infty$) tends to equalise the populations, i.e. $w \rightarrow 0$. Equivalently, the upper level has a steady-state population of

$$\rho_{22} = \frac{1 - w}{2} = \frac{\Omega^2/4}{\delta^2 + \Omega^2/2 + \Gamma^2/4}, \quad (7.69)$$

and $\rho_{22} \rightarrow 1/2$ as the intensity increases. This key result is used in Chapter 9 on radiation forces.

In the above, the optical Bloch equations have been justified by analogy with a damped classical oscillator but they also closely resemble the Bloch equations that describe the behaviour of a spin-1/2 particle in a combination of static and oscillating magnetic fields.³⁶ The reader familiar with magnetic resonance techniques may find it useful to make an analogy with that historically important case.³⁷ For times much shorter than any damping or relaxation time, the two-level atom and spin-1/2 system behave in the same way, i.e. they have a coherent evolution such as π - and $\pi/2$ -pulses, etc. A steady-state solution of the optical Bloch equations has been presented (and nothing has been said about the different result for a spin-1/2 system).

³⁴The main purpose of the rather lengthy discussion of the classical case was to highlight the correspondence between u , v and U , V to make this step seem reasonable. An auxiliary feature of this approach is to remind the reader of the classical electron oscillator model of absorption and dispersion (which is important in atomic physics).

³⁵The steady-state solution is obtained by setting $\dot{u} = \dot{v} = \dot{w} = 0$ in eqns 7.67 to give three simultaneous equations.

³⁶The Zeeman effect leads to a splitting between states with $m_s = \pm 1/2$ to give a two-level system, and the oscillating magnetic field drives *magnetic dipole* transitions between the levels. In atomic physics such transitions occur between Zeeman states and hyperfine levels (Chapter 6).

³⁷The Bloch equations were well known from magnetic resonance techniques before lasers allowed the observation of coherent phenomena in optical transitions. Radio-frequency transitions have negligible spontaneous emission and the magnetic dipole of the whole sample decays by other mechanisms. Where the optical Bloch equations (eqns 7.67) have decay constants of Γ and $\Gamma/2$ for the population and coherences, respectively, the Bloch equations have $1/T_1$ and $1/T_2$. The decay rates $1/T_1$ and $1/T_2$ in magnetic resonance techniques are expressed in terms of T_1 and T_2 , the longitudinal and transverse relaxation times, respectively. Under some conditions the two relaxation times are similar, but in other cases $T_2 \ll T_1$. T_1 describes the relaxation of the component of the magnetic moment parallel to the applied field $\mathbf{B}$ which requires exchange of energy (e.g. with the phonons in a solid). T_2 arises from the dephasing of individual magnetic moments (spins) so that the magnetisation of the sample perpendicular to $\mathbf{B}$ decays. For further details see condensed matter texts, e.g. Kittel (2004).

7.6 The optical absorption cross-section

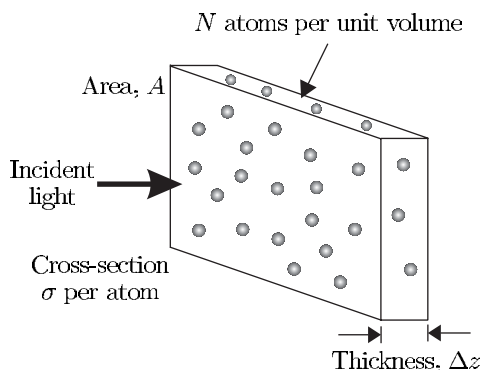
Monochromatic radiation causes an atom to undergo Rabi oscillations, but when the transition has damping the atom settles down to a steady state in which the excitation rate equals the decay rate. This has been shown explicitly above for an optical transition with spontaneous emission, but the same reduction of the coherent evolution of quantum amplitudes to a simple rate equation for populations (amplitudes squared) also occurs for other line-broadening mechanisms, e.g. Doppler broadening (Chapter 8) and collisions. Thus the equilibrium situation for monochromatic radiation is described by rate equations like those in Einstein's treatment of excitation by broadband radiation (eqns 1.25). It is convenient to write these rate equations in terms of an optical absorption cross-section defined in the usual way, as in Fig. 7.4. Consider a beam of particles (in this case photons) passing through a medium with N atoms per unit volume.³⁸ A slab of thickness Δz has $N\Delta z$ atoms per unit area and the fraction of particles absorbed by the target atoms is $N\sigma\Delta z$, where σ is defined as the cross-section; $N\sigma\Delta z$ gives the fraction of the target area covered by the atoms and this equals the probability that an incident particle hits an atom in the target (as it passes through the slab). The parameter σ that characterises the probability of absorption is equally well definable in quantum mechanics (in which photons and particles are delocalised, fuzzy objects) even though this cross-section generally has little relation to the physical size of the object (as we shall see). The probability of absorption equals the fraction of intensity lost, $\Delta I/I = -N\sigma\Delta z$, so the attenuation of the beam is described by

$$\frac{dI}{dz} = -\kappa(\omega) I = -N\sigma(\omega) I, \quad (7.70)$$

where $\kappa(\omega)$ is the absorption coefficient at the angular frequency ω of the incident photons. Integration gives an exponential decrease of the intensity with distance, namely

$$I(\omega, z) = I(\omega, 0) \exp \{-\kappa(\omega) z\}. \quad (7.71)$$

Fig. 7.4 Atoms with number density N distributed in a slab of thickness Δz absorb a fraction $N\sigma\Delta z$ of the incident beam intensity, where σ is defined as the cross-section (for absorption). $N\Delta z$ is the number of atoms per unit area and σ represents the 'target' area that each atom presents. We assume that the motion of the target atoms can be ignored (Doppler broadening is treated in Exercise 7.9) and also that atoms in the next layer (of thickness δz) cannot 'hide' behind these atoms (see Brooker 2003, Problem 3.26).



This formula, known as Beer's law (see Fox 2001), works well for absorption of low-intensity light that leaves most of the population in the ground state. Intense laser light significantly affects the populations of the atomic levels and we must take this into account. Atoms in level 2 undergo stimulated emission and this process leads to a gain in intensity (amplification) that offsets some of the absorption. Equation 7.70 must be modified to³⁹

$$\frac{dI}{dz} = -\kappa(\omega) I(\omega) = -(N_1 - N_2)\sigma(\omega) I(\omega). \quad (7.72)$$

Absorption and stimulated emission have the same cross-section. For the specific case of a two-level atom this can be seen from the symmetry with respect to the exchange of the labels 1 and 2 in the treatment of the two-level atom in the early parts of this chapter; the oscillating electric field drives the transition from 1 to 2 at the same rate as the reverse process—only the spontaneous emission goes one way. This is an example of the general principle that a strong absorber is also a strong emitter.⁴⁰ This is also linked to the equality of the Einstein coefficients, $B_{12} = B_{21}$, for non-degenerate levels. The population densities in the two levels obey the conservation equation $N = N_1 + N_2$.⁴¹ In the steady state conservation of energy per unit volume of the absorber requires that

$$(N_1 - N_2)\sigma(\omega) I(\omega) = N_2 A_{21} \hbar \omega. \quad (7.73)$$

On the left-hand side is the amount by which the rate of absorption of energy exceeds the stimulated emission, i.e. the net rate of energy absorbed per unit volume. On the right-hand side is the rate at which the atoms scatter energy out of the beam—the rate of spontaneous emission for atoms in the excited state times $\hbar\omega$.⁴² The number densities are related to the variables in the optical Bloch equations by $\rho_{22} = N_2/N$ and

$$w = \frac{N_2 - N_1}{N}, \quad (7.74)$$

and w and ρ_{22} are given in eqns 7.68 and 7.69, respectively. Hence

$$\sigma(\omega) = \frac{\rho_{22}}{w} \frac{A_{21} \hbar \omega}{I} = \frac{\Omega^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4} \times \frac{A_{21} \hbar \omega}{I}. \quad (7.75)$$

Both I and Ω^2 are proportional to $|E_0|^2$ so this cancels out, and further manipulation yields⁴³

$$\sigma(\omega) = 3 \times \frac{\pi^2 c^2}{\omega_0^2} A_{21} g_H(\omega). \quad (7.76)$$

The Lorentzian frequency dependence is expressed by the line shape function

$$g_H(\omega) = \frac{1}{2\pi} \frac{\Gamma}{(\omega - \omega_0)^2 + \Gamma^2/4}, \quad (7.77)$$

where the subscript H denotes homogeneous, i.e. something that is the same for each atom, like the radiative broadening considered here.⁴⁴ The

³⁹We do not try to include the degeneracy of the levels because illumination with intense polarized laser light usually leads to *unequal* populations of the states with different M_J , or M_F . This differs from the usual situation in laser physics where the excitation, or pumping mechanisms, populate all states in a given level at the same rate, so N_1/g_1 and N_2/g_2 can be taken as the population densities per state. Selective excitation of the upper level can give $N_2/g_2 > N_1/g_1$, and hence gain.

⁴⁰The laws of thermodynamics require that an object stays in equilibrium with black-body radiation at the same temperature, hence the absorbed and emitted powers must balance.

⁴¹Compare this with eqns 1.26, 7.7 and 7.43.

⁴²This assumes that atoms do not get rid of their energy in any other way such as inelastic collisions.

⁴³Intensity is related to the electric field amplitude by $I = \epsilon_0 c |E_0(\omega)|^2/2$, and $\Omega^2 = e^2 X_{12}^2 |E_0|^2/\hbar^2$ (eqn 7.12). Also $X_{12}^2 = |D_{12}|^2/3 \propto A_{21}$ (eqn 7.23). The degeneracy factors are $g_1 = g_2 = 1$ for the two-level atom, but see eqn 7.79.

⁴⁴It is a general result that homogeneous broadening mechanisms give a Lorentzian line shape.

area under this line shape function equals unity:

$$\int_{-\infty}^{\infty} g_H(\omega) d\omega = 1. \quad (7.78)$$

The pre-factor of 3 in eqn 7.76 may have any value in the range 0 to 3. It has the maximum value of 3 for atoms with the optimum orientation to absorb a beam of polarized laser light (from a specific direction). However, if either the light is unpolarized or the atoms have a random orientation (i.e. they are uniformly distributed across all the M_J states or M_F states) then the pre-factor is 1 because $|X_{12}|^2 = |D_{12}|^2/3$ as in eqn 7.21 (from the average of $\cos^2 \theta$ over all angles), and this 1/3 cancels the pre-factor of 3.⁴⁵ Under these conditions the absorption does not depend on the magnetic state (M_J or M_F) so a real atom with degenerate levels has a cross-section of

$$\sigma(\omega) = \frac{g_2}{g_1} \times \frac{\pi^2 c^2}{\omega_0^2} A_{21} g_H(\omega). \quad (7.79)$$

This equation, or eqn 7.76, applies to many experimental situations. Careful study of the following examples gives physical insight that can be applied to other situations.⁴⁶

Example 7.2 *Atoms in a specific M_F state interacting with a polarized laser beam, e.g. sodium atoms in a magnetic trap that absorb a circularly-polarized probe beam (Fig. 7.5)*

This gives effectively a two-level system and the polarization of the light matches the atom's orientation so eqn 7.76 applies (the pre-factor has the maximum value of 3). To drive the $\Delta M_F = +1$ transition the

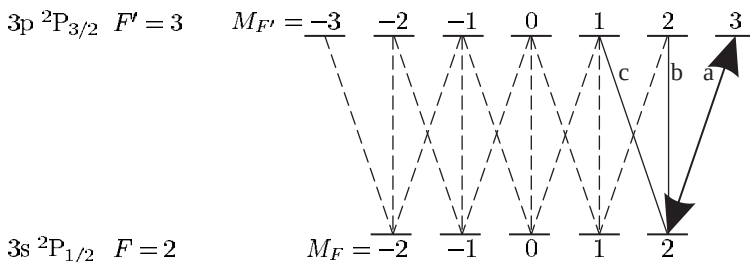


Fig. 7.5 The Zeeman states of the $3s \ ^2P_{1/2} \ F = 2$ and $3p \ ^2P_{3/2} \ F' = 3$ hyperfine levels of sodium, and the allowed electric dipole transitions between them. The other hyperfine levels ($F = 1$ and $F' = 0, 1$ and 2) have not been shown. Excitation of the transition $F = 2, M_F = 2$ to $F' = 3, M_{F'} = 3$ (labelled a) gives a closed cycle that has similar properties to a two-level atom—the selection rules dictate that atoms in the $F' = 3, M_{F'} = 3$ state spontaneously decay back to the initial state. (Circularly-polarized light that excites $\Delta M_F = +1$ transitions leads to cycles of absorption and emission that tend to drive the population in the $F = 2$ level towards the state of maximum M_F , and this optical pumping process provides a way of preparing a sample of atoms in this state.) When all the atoms have the correct orientation, i.e. they are in the $F = 2, M_F = 2$ state for this example, then eqn 7.76 applies. Atoms in this state give less absorption for linearly-polarized light (transition b), or circular polarization of the wrong handedness (transition c).

⁴⁵Spontaneously emitted photons go in random directions so an average over angles always occurs in the calculation of A_{21} .

⁴⁶Typically, for atoms with a well-defined orientation the polarization of the light is chosen to give the maximum cross-section. If this is not the case then the angular momentum algebra may be required to calculate the matrix elements. Only in special cases would the polarization be chosen to give a very weak interaction, i.e. a pre-factor much less than unity in eqn 7.76.

circularly-polarized light must have the correct handedness and propagate along the atom's quantisation axis (defined by the magnetic field in this example).⁴⁷

Example 7.3 *The absorption of light on an s-p transition, e.g. the 3s-3p resonance line of sodium*

Light with a particular polarization and direction drives a transition to one magnetic sub-level in the upper level, as shown in Fig. 7.6(a). Since the lower level has only $m_l = 0$ there is no averaging over the orientation and eqn 7.76 applies. Unpolarized light drives transitions to the three upper m_l states equally. For each transition the averaging gives a factor of $1/3$ but all three transitions contribute equally to the absorption so the atoms have the same cross-section as for polarized light (eqn 7.79 with $g_2/g_1 = 3$). Thus the s-p transition is a special case that gives the same absorption cross-section whatever the polarization of the light. Atoms with $m_l = 0$ have no preferred direction and interact in the same way with light of any polarization (or direction). In contrast, for the p-s transition shown in Fig. 7.6(b), atoms in a given m_l state only interact with light that has the correct polarization to drive the transition to $m_l = 0$.

⁴⁷The direction of the electric field at the atom depends on both the polarization *and* direction of the radiation, e.g. circularly-polarized light that propagates perpendicular to the quantisation axis drives $\Delta M_F = 0$ and ± 1 (π - and σ -) transitions. This leads to a smaller cross-section than when the light propagates along the axis. Radiation that propagates in all directions does not produce a polarized electric field, e.g. isotropic radiation in a black-body enclosure.

7.6.1 Cross-section for pure radiative broadening

The peak absorption cross-section given by eqn 7.76, when $\omega = \omega_0$, is

$$\sigma(\omega_0) = 3 \times \frac{2\pi c^2}{\omega_0^2} \frac{A_{21}}{\Gamma}. \quad (7.80)$$

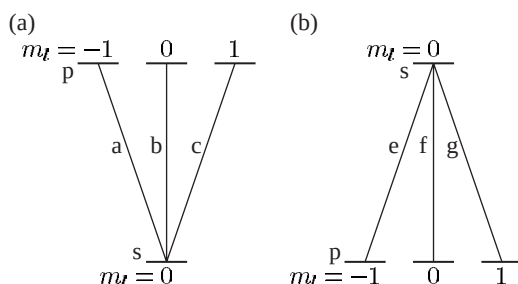


Fig. 7.6 A comparison of s-p and p-s transitions. (a) The three transitions a, b and c between the s and p levels have equal strength. The physical reason for this is that the spontaneous decay rate of the upper m_l states cannot depend on the atom's orientation in space. Light linearly-polarized parallel to the z -axis drives π -transition b only, and spontaneous decay occurs back to the initial state since there are no other accessible states—this gives the equivalent of a two-level system. The s-p transition is a special case where the absorption does not depend on the polarization, e.g. unpolarized light gives equal excitation rates on the three transitions a, b and c, and this increases the absorption by the degeneracy factor $g_2/g_1 = 3$, thereby cancelling the $1/3$ that arises in the orientational average. (b) In contrast, for the p-s transition the peak cross-section is one-ninth of that in (a).⁴⁸

⁴⁸Spin is ignored here. This applies when either the fine structure is not resolved, e.g. this may arise for the transition $2s-3p$ in hydrogen where the fine structure of the upper level is small, or to transitions between singlet terms, i.e. $^1S-^1P$ and $^1P-^1S$ (with $m_l \rightarrow M_l$ in the figure).

In a two-level atom the upper level can only decay to level 1 so $\Gamma = A_{21}$, and for a transition of wavelength $\lambda_0 = 2\pi c/\omega_0$ we find

$$\sigma(\omega_0) = 3 \times \frac{\lambda_0^2}{2\pi} \simeq \frac{\lambda_0^2}{2}. \quad (7.81)$$

This maximum cross-section is much larger than the size of the atom, e.g. the $\lambda_0 = 589 \text{ nm}$ transition of sodium has $\sigma(\omega_0) = 2 \times 10^{-13} \text{ m}^2$, whereas in kinetic theory the atoms have a cross-section of only $\pi d^2 = 3 \times 10^{-18} \text{ m}^2$ for an atomic diameter of $d = 0.3 \text{ nm}$ —‘collisions’ between atoms and photons have a large resonant enhancement. The optical cross-section decreases rapidly off resonance, e.g. light of wavelength 600 nm gives $\Gamma/(\omega - \omega_0) = 10^{-6}$ for the sodium transition above, so that $\sigma(\omega) = 10^{-12} \times \sigma(\omega_0) = 2 \times 10^{-25} \text{ m}^2$. Clearly the absorption of radiation has little relation to the size of the electronic orbitals.

7.6.2 The saturation intensity

In the previous section we calculated the absorption cross-section starting from eqn 7.73 and we shall now use the same equation to determine the population difference; we can write eqn 7.73 as $(N_1 - N_2) \times r = N_2$, where the dimensionless ratio $r = \sigma(\omega) I(\omega) / (\hbar \omega A_{21})$. This equation and $N_1 + N_2 = N$ give the difference in population densities as

$$N_1 - N_2 = \frac{N}{1 + 2r} = \frac{N}{1 + I/I_s(\omega)}, \quad (7.82)$$

where the saturation intensity is defined by

$$I_s(\omega) = \frac{\hbar \omega A_{21}}{2\sigma(\omega)}. \quad (7.83)$$

It is important to note that other definitions of saturation intensity are also used, such as the above expression without 2 in the denominator. From eqn 7.72 we find that the absorption coefficient depends on intensity as follows:⁴⁹

$$\kappa(\omega, I) = \frac{N\sigma(\omega)}{1 + I/I_s(\omega)}. \quad (7.84)$$

The *minimum* value of $I_{\text{sat}}(\omega)$ occurs on resonance where the cross-section is largest; this minimum value is often referred to as *the* saturation intensity, $I_{\text{sat}} \equiv I_s(\omega_0)$, given by⁵⁰

$$I_{\text{sat}} = \frac{\pi}{3} \frac{\hbar c}{\lambda^3 \tau}, \quad (7.85)$$

where $\tau = \Gamma^{-1}$ is the lifetime for radiative broadening. For example, the resonance transition in sodium at $\lambda = 589 \text{ nm}$ has a lifetime of $\tau = 16 \text{ ns}$ and for an appropriate polarization (as in Fig. 7.5) the atom cycles on an effectively two-level transition. This leads to an intensity $I_{\text{sat}} = 60 \text{ W m}^{-2}$, or 6 mW cm^{-2} , that can easily be produced by a tunable dye laser.

⁴⁹This equation is very similar to the formula for the saturation of gain in a homogeneously-broadened laser system since gain is negative absorption.

⁵⁰Here $I_{\text{sat}} = \hbar \omega A_{21} / (2\sigma(\omega_0))$ and $\sigma(\omega_0)$ is given by eqn 7.81.

We can also obtain eqn 7.82 directly from the steady-state value of $w = (N_1 - N_2)/N$ in eqn 7.68 if the saturation intensity is defined by

$$\frac{I}{I_{\text{sat}}} = \frac{2\Omega^2}{\Gamma^2}. \quad (7.86)$$

This is equivalent to eqn 7.85.⁵¹ At saturation the Rabi frequency has a value comparable with Γ .

⁵¹As discussed after eqn 7.75, Ω^2/I does not depend on the electric field.

7.6.3 Power broadening

Equation 7.84 for $\kappa(\omega, I)$ contains two quantities that vary with frequency: $\sigma(\omega)$ and $I_s(\omega)$.⁵² Rearranging this equation to show the frequency dependence, and defining $\sigma_0 \equiv \sigma(\omega_0)$ as the maximum cross-section in eqn 7.81, we find that

$$\begin{aligned} \kappa(\omega, I) &= N\sigma_0 \frac{\Gamma^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4} \times \frac{1}{1 + \frac{I}{I_{\text{sat}}} \frac{\Gamma^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4}} \\ &= N\sigma_0 \frac{\Gamma^2/4}{(\omega - \omega_0)^2 + \frac{1}{4}\Gamma^2(1 + I/I_{\text{sat}})}. \end{aligned} \quad (7.87)$$

The expression for the absorption coefficient $\kappa(\omega, I)$ has a Lorentzian line shape with a full width at half maximum (FWHM) of

$$\Delta\omega_{\text{FWHM}} = \Gamma \left(1 + \frac{I}{I_{\text{sat}}}\right)^{1/2}. \quad (7.88)$$

The line width increases with intensity. This **power broadening** occurs because saturation reduces the absorption near the resonance while far from resonance the absorption changes little (see Fig. 7.7). The expression for the population in the upper level ρ_{22} in eqn 7.69 also has the same power-broadened line width, as in eqn 7.88.⁵³ The relationship between this absorption and the populations of the two levels is discussed in Exercise 7.11.

⁵²In terms of the *minimum* value $I_{\text{sat}} = I_s(\omega_0)$ we can rewrite eqn 7.83 as

$$\frac{I_s(\omega)}{I_{\text{sat}}} = \frac{\sigma_0}{\sigma(\omega)}.$$

We also have

$$\sigma(\omega) = \sigma_0 \frac{\Gamma^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4}.$$

⁵³This can be shown by rearrangement of the denominator of eqn 7.69 in terms of a line width

$$\begin{aligned} \Delta\omega &= \Gamma \left(1 + \frac{2\Omega^2}{\Gamma^2}\right)^{1/2} \\ &= \Gamma \left(1 + \frac{I}{I_{\text{sat}}}\right)^{1/2}. \end{aligned}$$

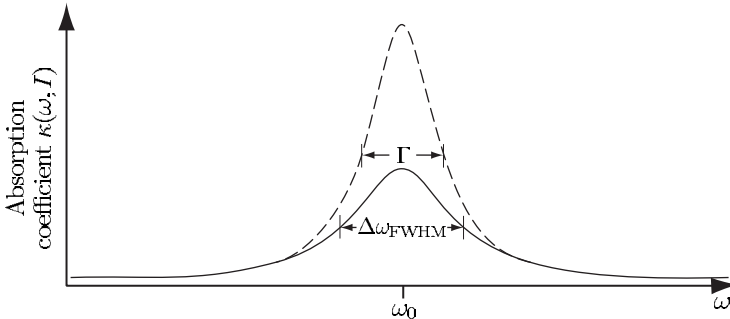


Fig. 7.7 The absorption coefficient $\kappa(\omega, I)$ is a Lorentzian function of the frequency that peaks at ω_0 , the atomic resonance. Saturation causes the absorption line shape to change from the curve for a low intensity ($I \ll I_{\text{sat}}$, dashed line), to a broader curve (solid line), with a lower peak value, described by the Lorentzian function in eqn 7.87.

7.7 The a.c. Stark effect or light shift

In addition to its effect on the populations, the perturbing radiation also changes the energy of the levels and we calculate this **light shift** in this section. We can write eqns 7.41 for $\tilde{c}_1$ and $\tilde{c}_2$ in matrix form as

$$i \frac{d}{dt} \begin{pmatrix} \tilde{c}_1 \\ \tilde{c}_2 \end{pmatrix} = \begin{pmatrix} \delta/2 & \Omega/2 \\ \Omega/2 & -\delta/2 \end{pmatrix} \begin{pmatrix} \tilde{c}_1 \\ \tilde{c}_2 \end{pmatrix}. \quad (7.89)$$

This has solutions of the form

$$\begin{pmatrix} \tilde{c}_1 \\ \tilde{c}_2 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} e^{-i\lambda t}. \quad (7.90)$$

The equation for the eigenvalues λ is

$$\begin{vmatrix} \delta/2 - \lambda & \Omega/2 \\ \Omega/2 & -\delta/2 - \lambda \end{vmatrix} = \lambda^2 - \left(\frac{\delta}{2}\right)^2 - \left(\frac{\Omega}{2}\right)^2 = 0. \quad (7.91)$$

Hence $\lambda = \pm (\delta^2 + \Omega^2)^{1/2}/2$. For $\Omega = 0$ the unperturbed eigenvalues are $\lambda = \pm \delta/2$, corresponding to two levels δ apart, as shown in Fig. 7.8. This result of time-dependent perturbation theory closely resembles the equations for a time-independent perturbation of two states with an energy separation of δ (see Appendix A); the two states are the excited state at E_2 and a level at energy $E_1 + \hbar\omega$, corresponding to the ground state plus a photon of the radiation field, see Fig. 7.9. This system of atom plus photon is called a ‘dressed atom’—see Cohen-Tannoudji *et al.* (1992) for a much more in-depth treatment.

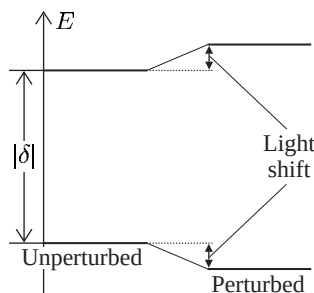
Normally light shifts are most important at large frequency detuning where the effect of absorption is negligible; in this case $|\delta| \gg \Omega$ and the eigenvalues are

$$\lambda \simeq \pm \left(\frac{\delta}{2} + \frac{\Omega^2}{4\delta} \right). \quad (7.92)$$

The states are shifted from their unperturbed eigenfrequencies by the light shift $\pm \Omega^2/4\delta$. From eqn 7.89 we see that the amplitude $\tilde{c}_1$ is associated with the state with unperturbed energy $+\delta/2$ and lies above the other state when $\delta > 0$. This state with amplitude $\tilde{c}_1$ has a light shift of

$$\Delta\omega_{\text{light}} = \frac{\Omega^2}{4\delta}. \quad (7.93)$$

Fig. 7.8 The treatment of the interaction of a two-level atom with radiation by time-dependent perturbation theory leads to eqn 7.89, that looks similar to a time-independent perturbation (proportional to Ω) of two energy levels with an energy separation of δ . The light shift is the difference between the unperturbed energies and energy eigenvalues of the system when it is perturbed by the radiation.



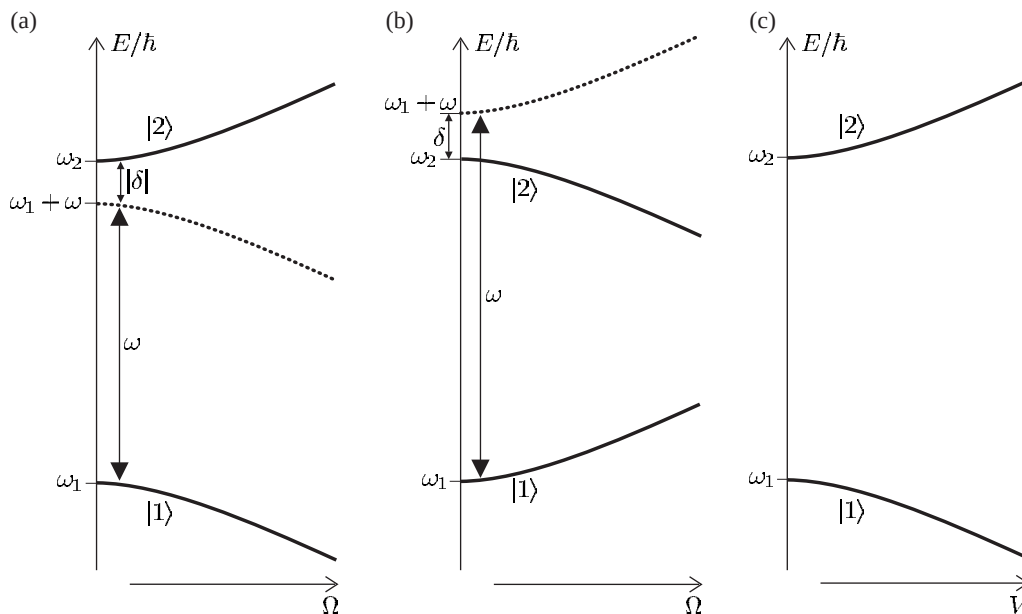


Fig. 7.9 Eigenenergies of a two-level atom interacting with an external electric field. (a) and (b) show the a.c. Stark effect for negative and positive frequency detunings respectively, as a function of the Rabi frequency. (c) The d.c. Stark effect as a function of the applied field strength.

This equation is also valid for negative frequency detuning $\delta < 0$ when the light shift of this state decreases its energy. The dependence of the light shift on the sign of δ has important consequences for dipole-force traps for atoms, as described in Chapter 9. Figure 7.9 summarises the light shift and shows the d.c. Stark shift for comparison, and the eigenstates of the perturbation are discussed in Appendix A.

7.8 Comment on semiclassical theory

This chapter's treatment of the interaction of radiation with atoms is semiclassical—the energy of the atoms is quantised but the radiation is not (since $\mathbf{E}_0 \cos \omega t$ is a classical electric field). It is individual two-level atoms that absorb energy in lumps of $\hbar\omega$ from the radiation; nevertheless, the quantity $I/\hbar\omega$ is commonly referred to as the flux of photons. In addition to transitions between two bound quantum states of the atom, this semiclassical theory can also describe photo-ionization where light excites an electron from a bound state to an unbound state above the ionization limit. In such a transition the atom suddenly becomes an ion (plus a free electron) at a given time, like a quantum jump between bound states. The average of many such jumps corresponds to the rate predicted semiclassically.⁵⁴ Photo-ionization of individual atoms closely resembles the photoelectric effect that occurs at the surface of a metal with work function Φ illuminated by light of frequency ω . The sur-

⁵⁴Similarly, in radioactivity the individual nuclei decay randomly but a large sample exhibits a smooth exponential decay.

face only emits electrons if $\hbar\omega > \Phi$. Semiclassical theory explains this observation if the least tightly bound energy level, or energy band, of electrons in the metal has a binding energy of Φ (cf. the ionization energy of atoms). Electrons come off the surface with a maximum kinetic energy of $\hbar\omega - \Phi$ because the oscillating field resonantly drives transitions that have angular frequencies close to ω (see eqn 7.15), from a lower level in the metal to an upper unbound ‘level’; this explanation does not require quantisation of the light into photons, as often implied in elementary quantum physics. A common line of argument is that a purely classical theory cannot explain various aspects of the photoelectric effect and therefore the light must be quantised. The above discussion shows that this effect can be explained by quantisation of the atoms.⁵⁵

⁵⁵The prompt emission of electrons after the light hits the surface can also be explained semiclassically.

However, there are phenomena that do require quantisation of the radiation field, e.g. in the fluorescence from a single trapped ion it is found that two photons have a lower probability of arriving at the detector together (within the short time period of a measurement) than predicted for a totally random source of photons. This spreading out of the photons, or *anti-bunching*, occurs because it takes a time to excite the ion again after spontaneous emission. Such correlation of photons, or anticorrelation in this case, goes beyond the semiclassical theory presented here. Quantitative calculations of photon statistics require a fully quantum theory called quantum optics. The book on this subject by Loudon (2000) contains more fascinating examples, and also gives rigorous derivations of many results used in this chapter.

7.9 Conclusions

The treatment of the interaction of atoms with radiation that has been described in this chapter forms the foundation of spectroscopy, e.g. laser spectroscopy as described in Chapter 8 (and laser physics); it is also important in quantum optics. There are a variety of approaches to this subject and it is worthwhile summarising the particular route that has been followed here.

The introductory sections closely follow Loudon (2000), and Section 7.3 gave a terse account of the coherent evolution of a two-level system interacting with single-frequency radiation—the atom undergoes Rabi oscillations and we saw that the Bloch sphere gives a useful way of thinking about the effect of sequences of π - and $\pi/2$ -pulses of the atom. In Section 7.5 the introduction of damping terms in the equations was justified by analogy with a behaviour of a classical dipole oscillator and this led to the optical Bloch equations. These equations give a complete description of the system and show how the behaviour at short times where damping has a negligible effect⁵⁶ is connected to what happens at longer times (much greater than the damping time) when a steady state has been established. We found that for radiative damping the system settles down to a steady-state solution described by a set of rate equations for the populations of the levels—this turns out to be a general

⁵⁶In the radio-frequency region the damping time can easily be longer than the time of measurement (even if it lasts several seconds). For allowed optical transitions the coherent evolution lasts for only a few nanoseconds, but it can be observed using short pulses of laser radiation. Optical experiments have also been carried out with exceptionally long-lived two-photon transitions (Demtröder 1996).

feature for all broadening mechanisms.⁵⁷ In particular, this means that the steady-state populations for an atom illuminated by monochromatic (laser) radiation can be related to the case of illumination by broadband radiation and the rate equations that Einstein wrote down. The theory of the interaction between radiation and matter can also be developed by working ‘backwards’ from the rate equations in Einstein’s treatment of broadband radiation to find the rate equations for the case of monochromatic radiation by making ‘reasonable’ assumptions about the atomic line shape, etc. This approach avoids perturbation theory and is commonly adopted in laser physics; however, it gives no information about coherent phenomena (Rabi oscillations, etc.).⁵⁸

Section 7.6 introduced the concept of an absorption cross-section and its use in the calculation of the absorption of radiation propagating through a gas with a certain number density of atoms.⁵⁹ The discussion of the cross-section provides a link between two different perspectives on the interaction of radiation with matter, namely (a) the effect of the radiation on the individual atoms, and (b) the effect of the atomic gas (medium) on the radiation, e.g. absorption.⁶⁰ The saturation of absorption as characterised by the saturation intensity forms the basis for a method of Doppler-free spectroscopy described in Chapter 8. From viewpoint (a), saturation arises because there is a maximum rate at which an atom can scatter radiation and this result for individual atoms is important in the discussion of radiation forces in Chapter 9. The formalism developed in this chapter allowed a straightforward derivation of the a.c. Stark effect on the atomic energy levels; this light shift is used in some of the methods of trapping and cooling atoms with laser radiation (also described in Chapter 9).

Further reading

Loudon’s book on quantum optics gives more rigorous derivations of many formulae in this chapter.⁶¹ Further properties of the optical Bloch equations are discussed by Cohen-Tannoudji *et al.* (1992) and Barnett and Radmore (1997). The treatment of the optical absorption cross-section of a gas closely resembles the calculation of the gain cross-section for a laser and further details can be found in books on laser physics including detailed discussion of broadening mechanisms, e.g. Davis (1996) and Corney (2000).

⁵⁷Both for mechanisms that are homogeneous like radiative broadening, e.g. collisions as shown in Loudon (2000), and also for inhomogeneous mechanisms such as Doppler broadening (Exercise 7.9).

⁵⁸Using time-dependent perturbation theory to describe the underlying behaviour of the two-level atom and finding the rate equations from them gives a clear understanding of the conditions under which these rate equations for the populations are valid.

⁵⁹The inverse of absorption is gain; this is a critical parameter in laser systems that is calculated in terms of an optical cross-section in a very similar way to absorption. The gain also exhibits saturation.

⁶⁰An important objective of this chapter was to show that both of these viewpoints embody the same physics.

⁶¹I have used similar notation, except Γ for the full width at half maximum (FWHM), whereas Loudon uses the half width $\gamma = \Gamma/2$.

Exercises

(7.1) *Averaging over spatial orientations of the atom*

- Light linearly polarized along the x -axis gives a dipole matrix element of $X_{12} = \langle 2|r|1 \rangle \cos \phi \sin \theta$. Show that the average over all solid angles gives a factor of $1/3$, as in eqn 7.21.
- Either show explicitly that the same factor of $1/3$ arises for light linearly polarized along the z -axis, $\mathbf{E} = E_0 \hat{\mathbf{e}}_z \cos \omega t$, or prove this by a general argument.

(7.2) *Rabi oscillations*

- Prove that eqns 7.25 lead to eqn 7.26 and that this second-order differential equation has a solution consistent with eqn 7.27.
- Plot $|c_2(t)|^2$ for the cases of $\omega - \omega_0 = 0$, Ω and 3Ω .

(7.3) *π - and $\pi/2$ -pulses*

- For zero detuning, $\omega = \omega_0$, and the initial conditions $c_1(0) = 1$ and $c_2(0) = 0$, solve eqns 7.25 to find both $c_1(t)$ and $c_2(t)$.
- Prove that a π -pulse gives the operation in eqn 7.30.
- What is the overall effect of two π -pulses acting on $|1\rangle$?
- Show that a $\pi/2$ -pulse gives $|1\rangle \rightarrow \{|1\rangle - i|2\rangle\}/\sqrt{2}$.
- What is the overall effect of two $\pi/2$ -pulses acting on $|1\rangle$? When state $|2\rangle$ experiences a phase shift of ϕ , between the two pulses, show that the probabilities of ending up in states $|1\rangle$ and $|2\rangle$ are $\sin^2(\phi/2)$ and $\cos^2(\phi/2)$, respectively.
- Calculate the effect of the three-pulse sequence $\pi/2 - \pi - \pi/2$, with a phase shift of ϕ between the second and third pulses. (The operators can be written as 2×2 unitary matrices, although this is not really necessary for this simple case.)

Comment. Without the factors of $-i$ the signals in the two output ports of the interferometer are not complementary. The fact that the identity operation is a 4π -pulse rather than 2π stems from

the isomorphism between the two-level atom and a spin-1/2 system.

(7.4) *The steady-state excitation rate with radiative broadening*

An alternative treatment of radiative decay simply introduces decay terms into eqn 7.25 to give

$$i\dot{c}_1 = c_2 \frac{\Omega}{2} e^{i(\omega - \omega_0)t} + i\frac{\Gamma}{2} c_2, \quad (7.94)$$

$$i\dot{c}_2 = c_1 \frac{\Omega}{2} e^{-i(\omega - \omega_0)t} - i\frac{\Gamma}{2} c_2. \quad (7.95)$$

This is a phenomenological model, i.e. a guess that works. The integrating factor $\exp(\Gamma t/2)$ allows eqn 7.95 to be written as

$$\begin{aligned} \frac{d}{dt} \left\{ c_2 \exp\left(\frac{\Gamma t}{2}\right) \right\} \\ = -ic_1 \frac{\Omega^*}{2} \exp\left\{ -i\left(\omega - \omega_0 + \frac{i\Gamma}{2}\right)t \right\}. \end{aligned}$$

- Show that for $\Omega = 0$ eqn 7.95 predicts that

$$|c_2(t)|^2 = |c_2(t=0)|^2 e^{-\Gamma t}.$$

- For the initial conditions $c_1(0) = 1$ and $c_2(0) = 0$, integration of eqn 7.94 gives $c_1 \simeq 1$. For these conditions and weak excitation ($\Omega \ll \Gamma$) show that, after a time which is long compared to the radiative lifetime, level 2 has a steady-state population given by

$$|c_2|^2 = \frac{\Omega^2/4}{(\omega - \omega_0)^2 + \Gamma^2/4}.$$

(7.5) *Saturation of absorption*

The 3s–3p resonance line of sodium has a wavelength of $\lambda = 589$ nm.

- Sodium atoms in a magnetic trap form a spherical cloud of diameter 1 mm. The Doppler shift and the Zeeman effect of the field are both small compared to Γ . Calculate the number of atoms that gives a transmission of $e^{-1} = 0.37$ for a weak resonant laser beam.
- Determine the absorption of a beam with intensity $I = I_{\text{sat}}$.

- (7.6) *The properties of some transitions in hydrogen*
The table gives the values of A_{21} for transitions from the $n = 3$ shell of hydrogen to lower levels. (Spin and fine structure are ignored.)

Transition	$A_{21} \text{ (s}^{-1}\text{)}$
2p–3s	6.3×10^6
1s–3p	1.7×10^8
2s–3p	2.2×10^7
2p–3d	6.5×10^7

- (a) Draw an energy-level diagram for the $n = 1, 2$ and 3 shells in hydrogen that shows the allowed electric dipole transitions between the orbital angular momentum levels. (Neglect transitions for which n does not change, e.g. 2s–2p.)
- (b) Calculate the lifetimes of the 3s, 3p and 3d configurations. What fraction of atoms that start in 3p end up in the 2s configuration?
- (c) An electron in the 2p configuration has a lifetime of only 1.6 ns. Why is this shorter than for the 3p configuration?
- (d) Calculate the radial matrix elements D_{12} in units of a_0 for the transitions in the table, and for 1s–2p.
- (e) Calculate I_{sat} for the 2p–3s and 1s–3p transitions.
- (7.7) *The classical model of atomic absorption*

- (a) A simple classical model of absorption assumes that an electron (in an atom) behaves like a damped simple harmonic oscillator of charge $-e$ and mass m_e driven by the oscillating electric field of the radiation: $E_0 \cos \omega t$. The electron's equation of motion has the form of eqn 7.55 with a driving force of constant amplitude $F_0 = -eE_0$. Find a solution of this equation in the form $x = \mathcal{U} \cos \omega t - \mathcal{V} \sin \omega t$ ($\mathcal{U}$ and $\mathcal{V}$ are not functions of time in the case considered here).
- (b) Show that the displacement of the electron has an amplitude

$$\begin{aligned} \sqrt{\mathcal{U}^2 + \mathcal{V}^2} &= \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (\beta\omega)^2}} \\ &\simeq \frac{F_0}{2m\omega} \left\{ (\omega - \omega_0)^2 + \frac{\beta^2}{4} \right\}^{-1/2}. \end{aligned}$$

Show that the angular frequency ω at which this amplitude is maximum is very close to ω_0 for a narrow resonance.

- (c) Show that the phase is given by

$$\tan \phi = \frac{\mathcal{V}}{\mathcal{U}} = \frac{\beta\omega}{\omega^2 - \omega_0^2}.$$

How does this phase vary as the angular frequency ω increases from $\omega \ll \omega_0$ to $\omega \gg \omega_0$?

- (d) For frequencies close to the atomic resonance ($\omega \simeq \omega_0$), show that your expressions for $\mathcal{U}$ and $\mathcal{V}$ can be written in an approximate form that agrees with eqns 7.58 and 7.59 (that were derived using the slowly-varying envelope approximation in which the amplitude of the driving force may change slowly over time).
- (e) Show that in the steady state the power P absorbed by the electron is a Lorentzian function of ω :

$$P \propto \frac{1}{(\omega - \omega_0)^2 + (\beta/2)^2}.$$

(7.8) *Oscillator strength*

This question shows the usefulness of a dimensionless parameter called the absorption oscillator strength, denoted by f_{12} .

- (a) Show that for the cross-section in eqn 7.79 we have

$$\int_{-\infty}^{\infty} \sigma(\omega) d\omega = 2\pi^2 r_0 c f_{12}, \quad (7.96)$$

where $r_0 = 2.8 \times 10^{-15} \text{ m}$ and $f_{12} = 2m_e \omega D_{12}^2 / (3\hbar)$.

- (b) From the simple model of the atom as an oscillating electron in Exercise 7.7, find the classical absorption cross-section $\sigma_{\text{cl}}(\omega)$ in terms of β , ω_0 and fundamental constants.
- (c) Without the driving electric field, the oscillator undergoes damped harmonic motion $x = x_0 e^{-\beta t/2} \cos(\omega' t - \varphi)$. The power radiated by an oscillating dipole leads to a decay rate given by eqn 1.23 (from classical electromagnetism). Determine β .
- (d) Show that $\sigma_{\text{cl}}(\omega)$ integrated over all frequencies gives $2\pi^2 r_0 c$.

Comment. This classical value is the maximum value for any transition, so $f_{12} \leq 1$. The absorption oscillator strength is a fraction of the integrated cross-section associated with a given transition.

- (e) Calculate f_{12} for the 3s–3p transition of sodium. ($A_{21} = \Gamma = 2\pi \times 10^7 \text{ s}^{-1}$.)
- (f) Calculate the absorption oscillator strength for the 1s–2p and 1s–3p transitions in hydrogen using the data from Exercise 7.6(d).

(7.9) *Doppler broadening*

The Maxwell–Boltzmann distribution of the velocities in a gas is a Gaussian function $f(v)$, as defined in eqn 8.3. Explain why for excitation by monochromatic radiation of angular frequency ω the population in the upper level is given by

$$|c_2(t)|^2 = \frac{e^2 X_{12}^2}{\hbar^2} |E(\omega)|^2 \times \int_{-\infty}^{\infty} \frac{\sin^2\{(\omega - \omega_0 + kv)t/2\}}{(\omega - \omega_0 + kv)^2} f(v) dv.$$

Assuming that the sinc^2 in the integrand acts like a Dirac delta function (as explained in Section 7.2), show that $|c_2|^2$ is proportional to $g_D(\omega)$ in eqn 8.4.

Comment. Doppler broadening washes out the Rabi oscillations because their frequency depends on the velocity, giving an equation similar to that for broadband radiation. For all broadening mechanisms Rabi oscillations, and other coherent phenomena, are only seen on time-scales shorter than the reciprocal of the line width.

(7.10) *An example of the use of Fourier transforms*

Show that an oscillator whose amplitude decays exponentially according to $x_0 e^{-\beta t/2} \cos(\omega t)$ radiates with a Lorentzian power spectrum.

(7.11) *The balance between absorption and spontaneous emission*

Explain why absorption and the population in the upper level are related by

$$\kappa(\omega, I) I = N_2 A_{21} \hbar \omega = N \rho_{22} A_{21} \hbar \omega. \quad (7.97)$$

Show that this is consistent with eqns 7.87 and 7.69 for $\kappa(\omega, I)$ and ρ_{22} , respectively.

Comment. An electric dipole does not radiate uniformly in all directions but this does not matter here; only a tiny fraction of the spontaneous emission goes along the direction of the incident beam. For example, in an experiment to measure the attenuation of a laser beam as it passes through a gas cell, a negligible fraction of the light scattered out of the beam falls on the photodetector that measures the power after the sample.

(7.12) *The d.c. Stark effect*

This exercise goes through a treatment of the d.c. Stark effect for comparison with the a.c. Stark effect.

- (a) An atom with two levels of energies $E_2 > E_1$, and a separation of $\epsilon = E_2 - E_1$ is placed in a *static* electric field. Show that the Hamiltonian for the system has the form

$$\hat{H} = \begin{pmatrix} \epsilon/2 & V \\ V & -\epsilon/2 \end{pmatrix},$$

where the matrix element for the perturbation V is proportional to the magnitude of the electric field. Find the energy eigenvalues. The two levels move apart as shown in Fig. 7.9—this is a general feature of systems where a perturbation mixes the wavefunctions.

- (b) Show that ‘weak’ fields produce a quadratic Stark effect on the atom, equivalent to the usual second-order perturbation-theory expression for a perturbation H_1 :

$$\Delta E_1 = -\frac{|\langle 2 | H_1 | 1 \rangle|^2}{E_2 - E_1}.$$

A similar expression can be found for the energy shift ΔE_2 of the other level (in the opposite direction).

- (c) Estimate the Stark shift for the ground state of a sodium atom in a field of 10^6 V m^{-1} (e.g. 10^4 V between plates 1 cm apart).

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.